KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4461 | 4 Credits: 4 | Program Core | Source-grounded

Strength of Materials (FS)

- To understand basic concepts of stress and strain in materials.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4461 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4461.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Fire Technology and Safety
Course code	4461
Course title	Strength of Materials (FS)
Semester	4 Credits: 4
Course category	Program Core
Periods per week	As specified in the official PDF
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To understand basic concepts of stress and strain in materials.
- ● To learn how to calculate shear force and bending moments in beams.
- ● To explore how beams bend and twist under load.
- ● To learn about torsion in shafts, spring deflection, and the stresses in thin cylinders.

Course outcomes

CO	Official outcome	Study level
CO1	Explain and apply the concepts specified in the official course syllabus.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 3
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand basic stress and strain and calculate changes in dimensions of	5	As specified
Module 2	cases.	3	As specified
Module 3	Understand the bending of beams, and solve problems related to deflection	5	As specified
Module 4	solve related problems.	3	As specified

MODULE 1

Understand basic stress and strain and calculate changes in dimensions of

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand basic stress and strain and calculate changes in dimensions of
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

2 Understanding and strain.

2 Understanding and strain. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and strain. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and strain. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding and strain. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding and strain. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and

Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand stress-strain diagrams for common 2 understanding materials. calculate stress and strain for uniform and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

2 Understanding composite sections. Understand thermal stresses in materials..

2 Understanding composite sections. Understand thermal stresses in materials.. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding composite sections. Understand thermal stresses in materials.. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding composite sections. understand thermal stresses in materials.. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding composite sections. Understand thermal stresses in materials.. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding composite sections. Understand thermal stresses in materials.. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding Solve problems on stress and strain.

2 Understanding Solve problems on stress and strain. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Solve problems on stress and strain. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding solve problems on stress and strain. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Solve problems on stress and strain. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding Solve problems on stress and strain. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple

5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying types of stress and strain: tensile, compressive, and shear. stress-strain diagrams for materials like mild steel (ms) and cast iron (ci).thermal stress in materials calculate shear force and bending moments, and draw diagrams for simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand basic stress and strain and calculate changes in dimensions of
            materials.

            Learn types of forces and basic concepts of stress
  M1.01                                                              2
Understanding
            and strain.
  M1.02     Understand stress-strain diagrams for common             2
Understanding
            materials.
            Calculate stress and strain for uniform and
  M1.03                                                              2
Understanding
            composite sections.
            Understand thermal stresses in materials..
  M1.04                                                              2
Understanding

            Solve problems on stress and strain.
  M1.05                                                              5
Applying

Contents:

      Types of stress and strain: tensile, compressive, and shear.
      Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron
(CI).Thermal
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

cases.

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cases.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment

Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand different types of beams and loads. 3 understanding learn shear force and bending moment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding definitions Calculate shear force and bending moments for

2 Understanding definitions Calculate shear force and bending moments for is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding definitions Calculate shear force and bending moments for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding definitions calculate shear force and bending moments for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding definitions Calculate shear force and bending moments for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 2 Understanding definitions Calculate shear force and bending moments for □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.

10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 10 applying various beam cases. types of beams: cantilever, simply supported, overhanging, etc. types of loads: point load, uniformly distributed load (udl). drawing shear force and bending moment diagrams. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
cases.

  M2.01     Understand different types of beams and loads.          3
Understanding

            Learn shear force and bending moment
  M2.02                                                             2
Understanding
            definitions
            Calculate shear force and bending moments for
  M2.03                                                             10
Applying
            various beam cases.
            Series Test 1                                        1

      Contents:

      Types of beams: cantilever, simply supported, overhanging, etc.
      Types of loads: point load, uniformly distributed load (UDL).
      Drawing shear force and bending moment diagrams.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Understand the bending of beams, and solve problems related to deflection

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the bending of beams, and solve problems related to deflection
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Understanding Understand the bending equation and apply it to

2 Understanding Understand the bending equation and apply it to is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Understand the bending equation and apply it to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding understand the bending equation and apply it to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Understand the bending equation and apply it to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding Understand the bending equation and apply it to □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding simple problems.

3 Understanding simple problems. is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding simple problems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding simple problems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding simple problems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding simple problems. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard

Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, calculate bending stress, modulus of section, and 6 applying moment of resistance. understand beam deflection and apply standard should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding formulae. Learn about columns and struts, and calculate

2 Understanding formulae. Learn about columns and struts, and calculate is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding formulae. Learn about columns and struts, and calculate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding formulae. learn about columns and struts, and calculate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding formulae. Learn about columns and struts, and calculate means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding formulae. Learn about columns and struts, and calculate □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler’s formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and

3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler’s formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler’s formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying buckling load. bending stress and moment of resistance. deflection of beams under point load and udl.columns: euler’s formula for buckling learn the behavior of shafts, springs, and thin cylinders under load, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler’s formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler’s formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand the bending of beams, and solve problems related to deflection
            and columns.

            Learn the basic theory of bending.
  M3.01                                                          2
Understanding


            Understand the bending equation and apply it to
  M3.02                                                          3
Understanding
            simple problems.
  M3.03  Calculate bending stress, modulus of section, and       6
Applying
         moment of resistance.
        Understand beam deflection and apply standard
  M3.04                                                          2
Understanding
        formulae.
            Learn about columns and struts, and calculate
  M3.05                                                          3
Applying
            buckling load.

Contents:


      Bending stress and moment of resistance.
      Deflection of beams under point load and UDL.Columns: Euler’s formula for
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

solve related problems.

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: solve related problems.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding of inertia (M.I.). Understand how springs work and calculate their

5 Understanding of inertia (M.I.). Understand how springs work and calculate their is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding of inertia (M.I.). Understand how springs work and calculate their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding of inertia (m.i.). understand how springs work and calculate their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding of inertia (M.I.). Understand how springs work and calculate their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 5 Understanding of inertia (M.I.). Understand how springs work and calculate their [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and

5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding deflection and stiffness. understand stresses in thin cylindrical shells and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and means in the official course context
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4 Understanding calculate their thickness. SeriesTest–II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press

4 Understanding calculate their thickness. SeriesTest–II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

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- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding calculate their thickness. seriestest–ii 1 torsion in shafts: calculation of polar m.i. and stress. closed coil helical springs: stress, deflection, and stiffness. stresses in thin cylindrical shells subjected to internal pressure. text / reference: t/r booktitle/author strength of materials by dr. r.k. bansal, lakshmi publishers t1 strength of materials by r.s. khurmi, s.chand& company ltd. t2 strength of materials by s.s. bhavikatti, vikas publishing house r1 r2 strength of materials by ramamrutham, dhanpat rai & sons r3 strength of materials by t.d. gunneswara rao, cambridge university press should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding calculate their thickness. SeriesTest–II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
solve related problems.

            Learn about shafts, calculate the polar moment
  M4.01                                                          5
Understanding
            of inertia (M.I.).
            Understand how springs work and calculate their
  M4.02                                                          5
Understanding
            deflection and stiffness.
            Understand stresses in thin cylindrical shells and
  M4.03                                                          4
Understanding
            calculate their thickness.
            SeriesTest–II                                        1
Contents:

        Torsion in shafts: calculation of polar M.I. and stress.
        Closed coil helical springs: stress, deflection, and stiffness.
        Stresses in thin cylindrical shells subjected to internal pressure.

Text / Reference:

  T/R                                     BookTitle/Author
          Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers
   T1
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding and strain.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding and strain.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and. (3 marks)

Answer: Begin with the standard definition or identity of *Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding composite sections. Understand thermal stresses in materials... (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding composite sections. Understand thermal stresses in materials..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding Solve problems on stress and strain.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Solve problems on stress and strain.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment. (3 marks)

Answer: Begin with the standard definition or identity of *Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding definitions Calculate shear force and bending moments for. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding definitions Calculate shear force and bending moments for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.. (3 marks)

Answer: Begin with the standard definition or identity of *10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding Understand the bending equation and apply it to. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Understand the bending equation and apply it to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding simple problems.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding simple problems.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard. (3 marks)

Answer: Begin with the standard definition or identity of *Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding formulae. Learn about columns and struts, and calculate. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding formulae. Learn about columns and struts, and calculate*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 5 Understanding of inertia (M.I.). Understand how springs work and calculate their. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding of inertia (M.I.). Understand how springs work and calculate their*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding calculate their thickness. SeriesTest–II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding calculate their thickness. SeriesTest–II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding and strain.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding Understand the bending equation and apply it to**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **5 Understanding of inertia (M.I.). Understand how springs work and calculate their**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson